

RECLAIM Project

DIGT 2107 - Project Iteration 3.1: Software Design Documentation

RECLAIM[Home](#)[Wellness ▼](#)[Support ▼](#)[Activities ▼](#)[Login](#)[Sign Up](#)

Reclaim Your Wellness

Track your mood, build healthy habits, and access support resources designed for university students.

How are you feeling today?

Track your daily mood to understand your wellness patterns



Great



Good



Okay



Not Good



Terrible

Add a note (optional)

What's on your mind today?

Save Mood Entry

Recent Mood Entries

Please [log in](#) to track your mood.

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1. INTRODUCTION

Course: DIGT2107 – Fall Term 2025

Instructor: Dr. May Haidar

Due Date: February 1, 2026

Project name: Reclaim

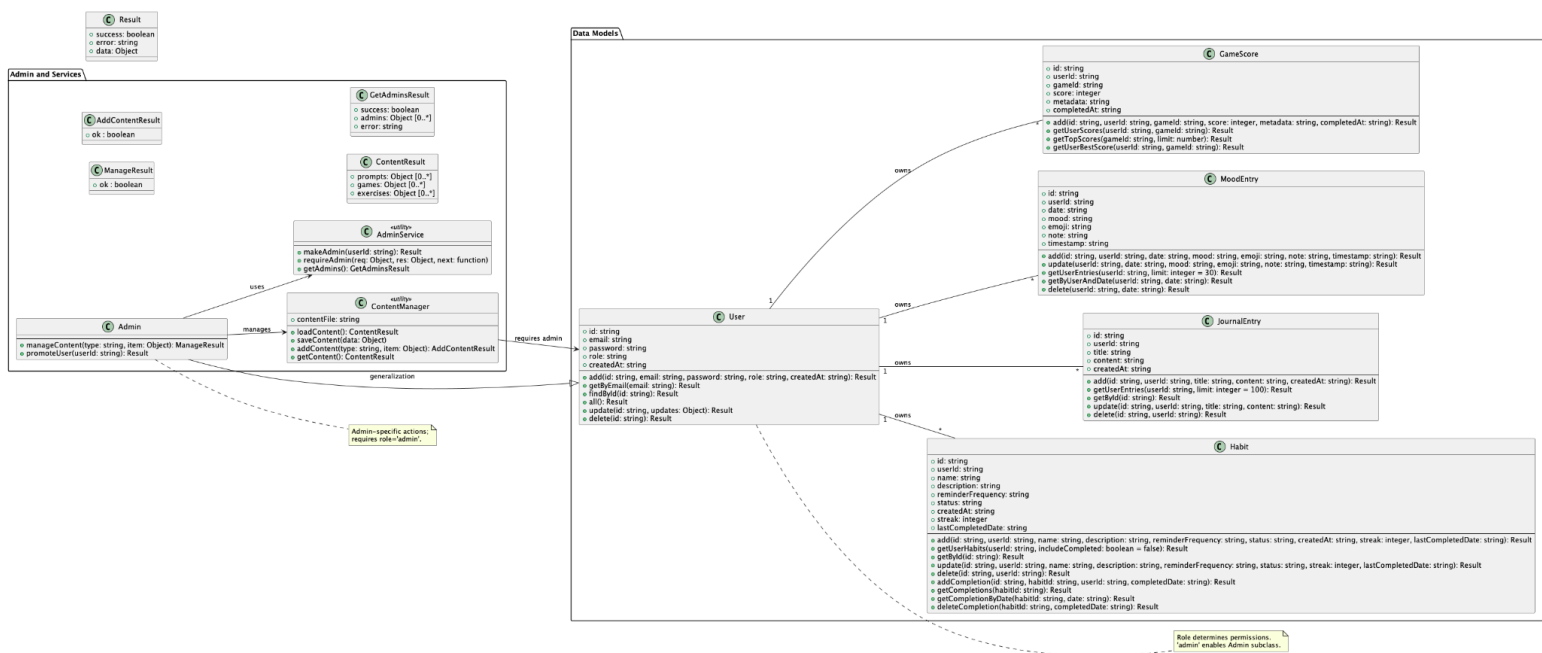
Team number: Group 4

Team members: Brianna Nguyen, Justin Gamez, Kieran Bendell, Ruqayya Naqvi

GitHub repository: <https://github.com/justingmz77/Reclaim>

2. SOFTWARE DESIGN DOCUMENTATION

2.1 Class Diagrams

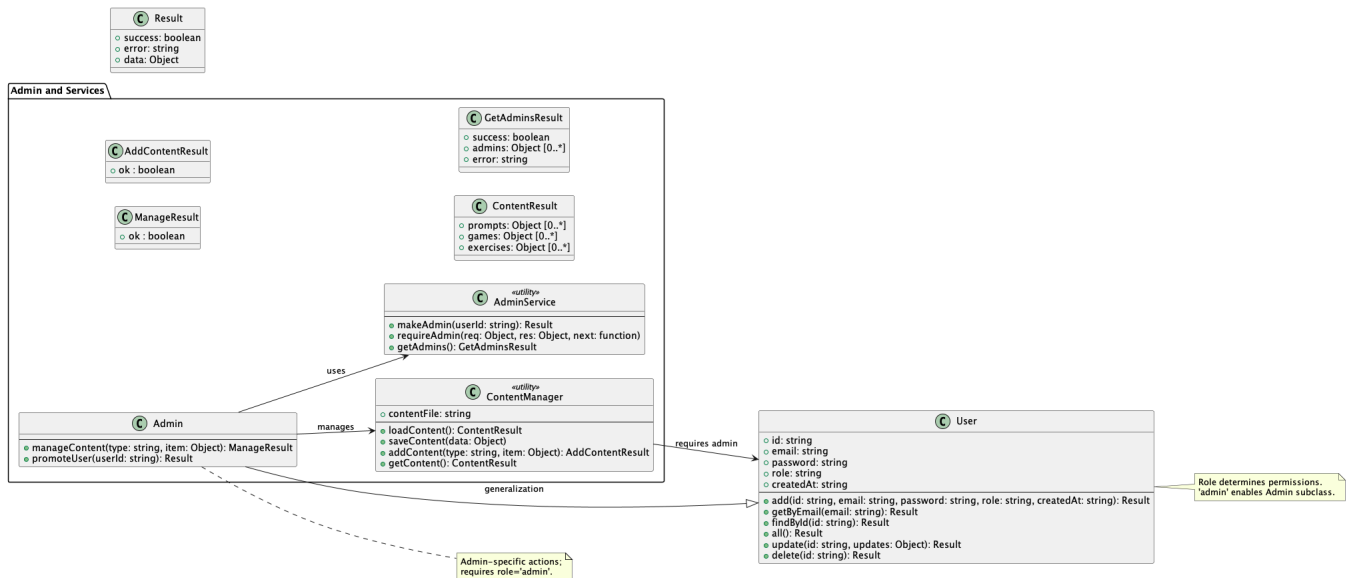


The class diagram is split into two smaller packages: Admin and Services and Data Models.

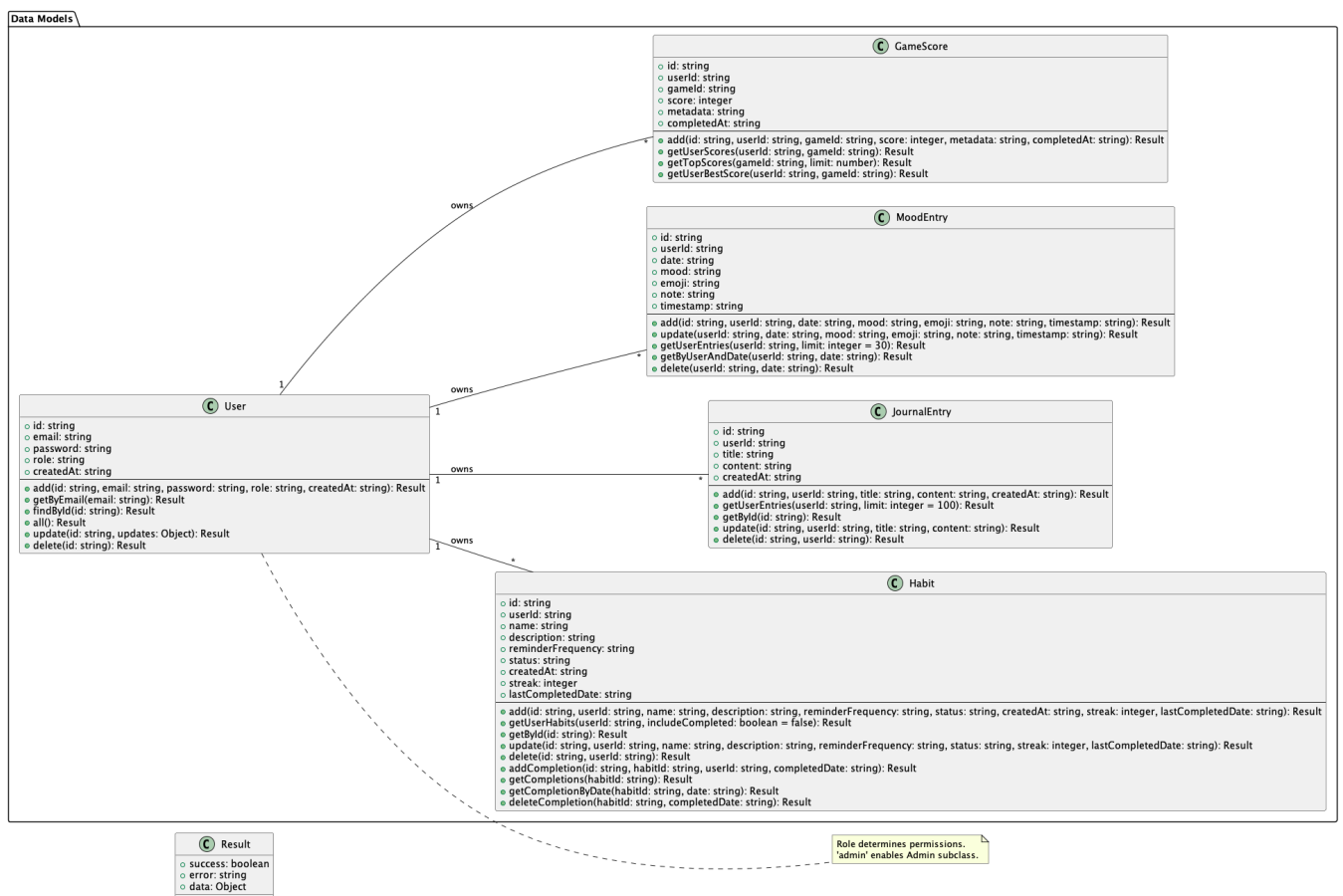
For many operations in the various classes, we return object literals. However, in UML, it is common practice to avoid using object literals to describe the return type of the operation. Therefore, there are multiple wrapper classes such as Result, ManagerResult, etc to depict the return type of many methods.

For better visualization, below are the two sections of the diagram separated.

Admin and Services package:

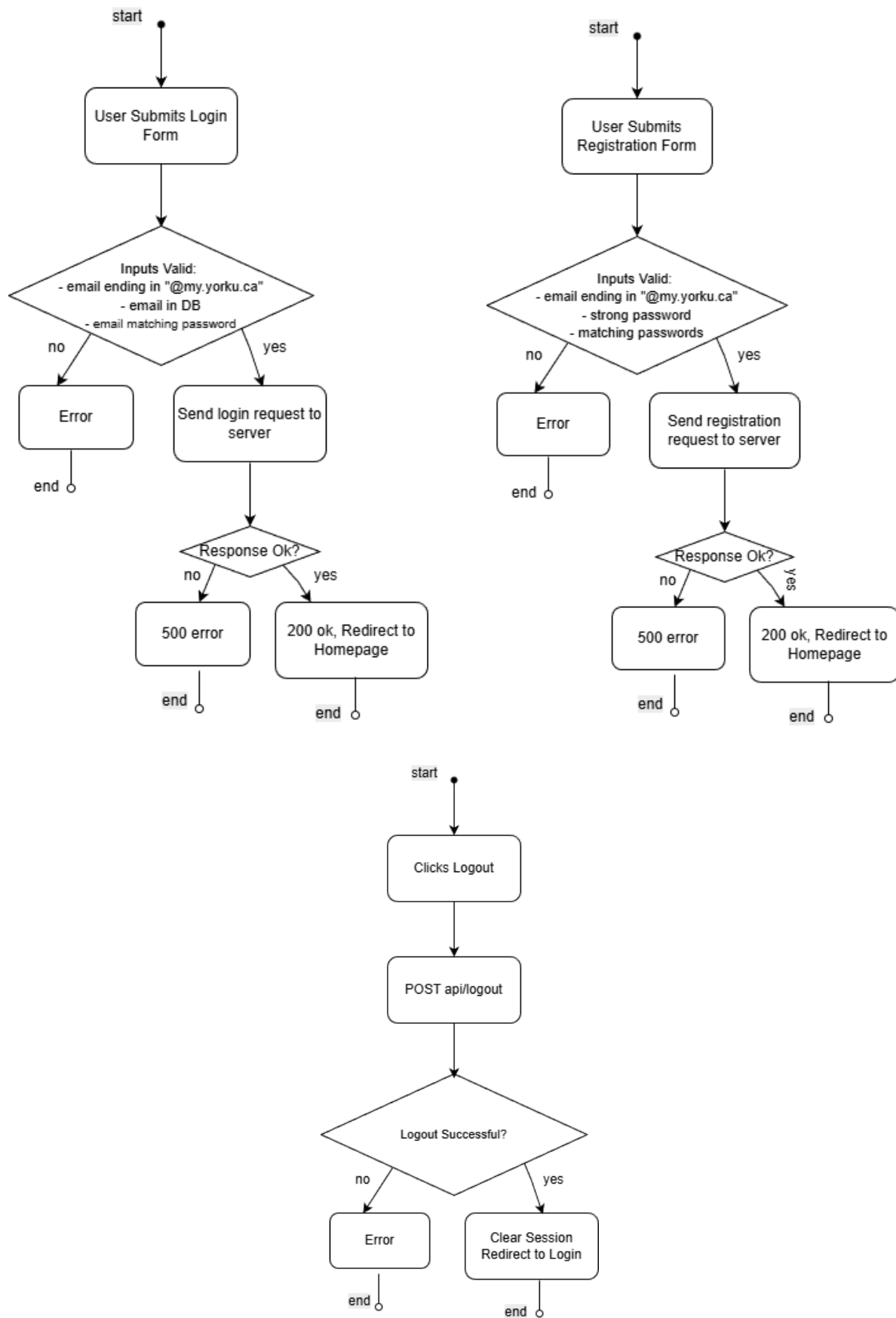


Data models package:

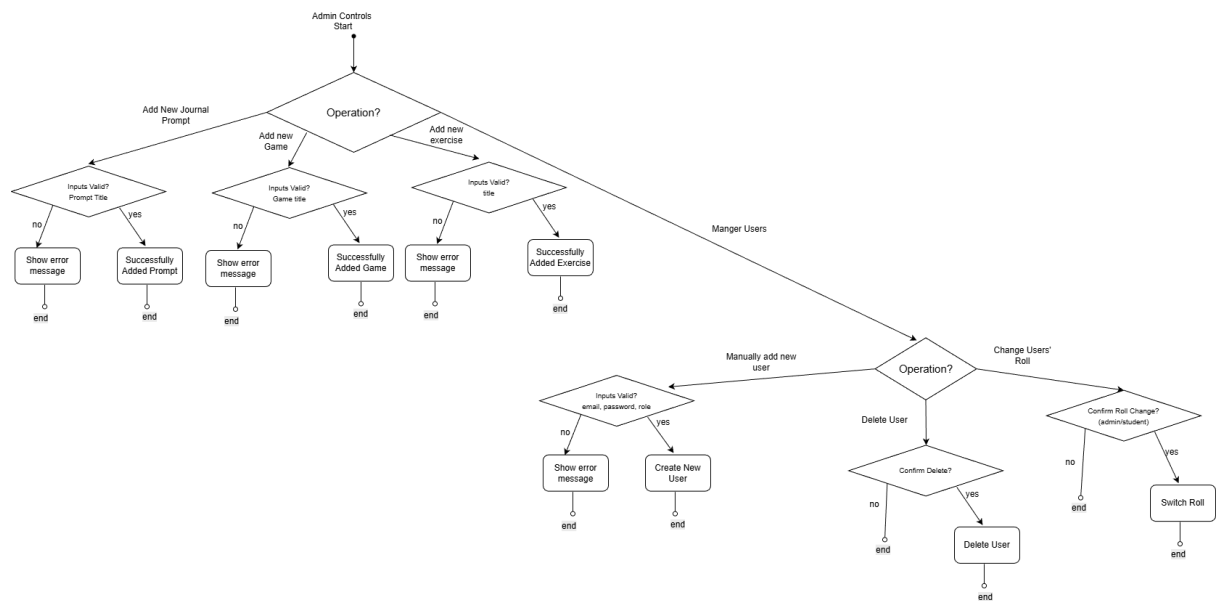
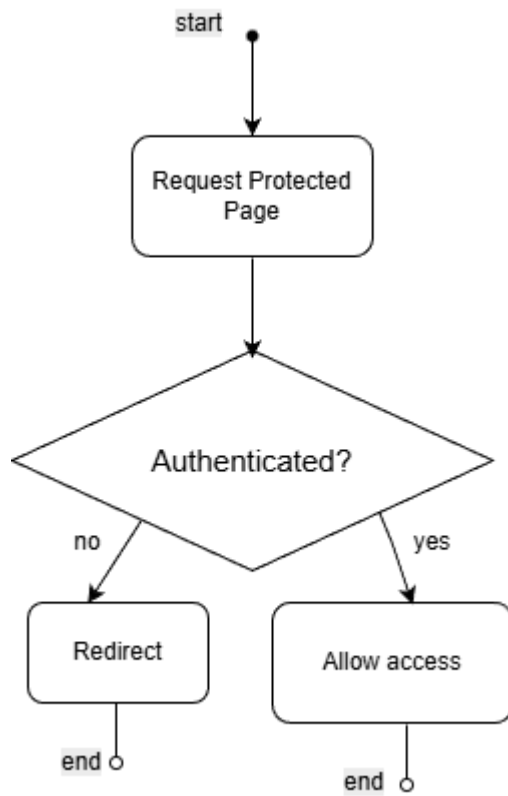


2.2 Activity Diagram

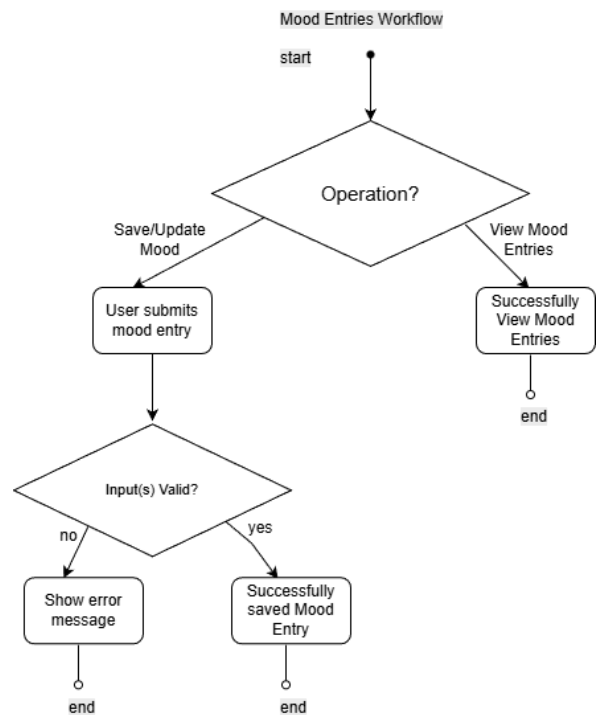
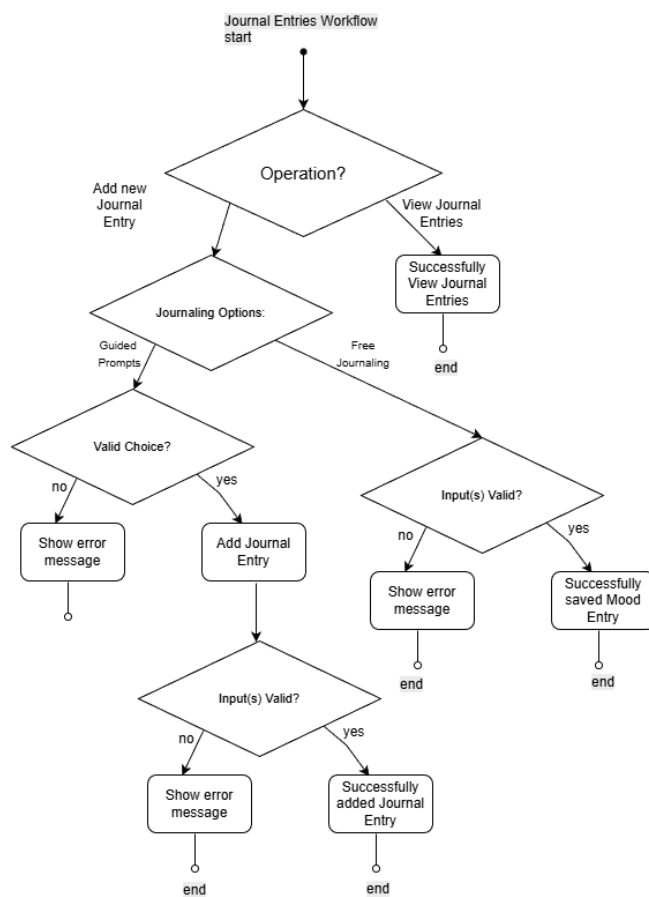
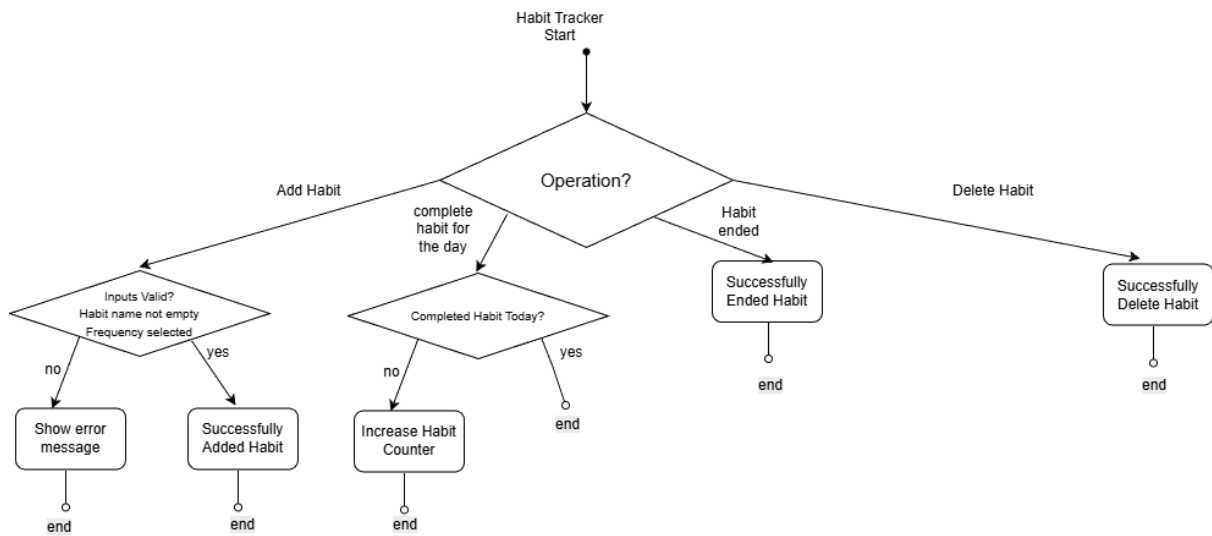
Registration, Login and Logout Diagrams:



Authentication Diagrams:

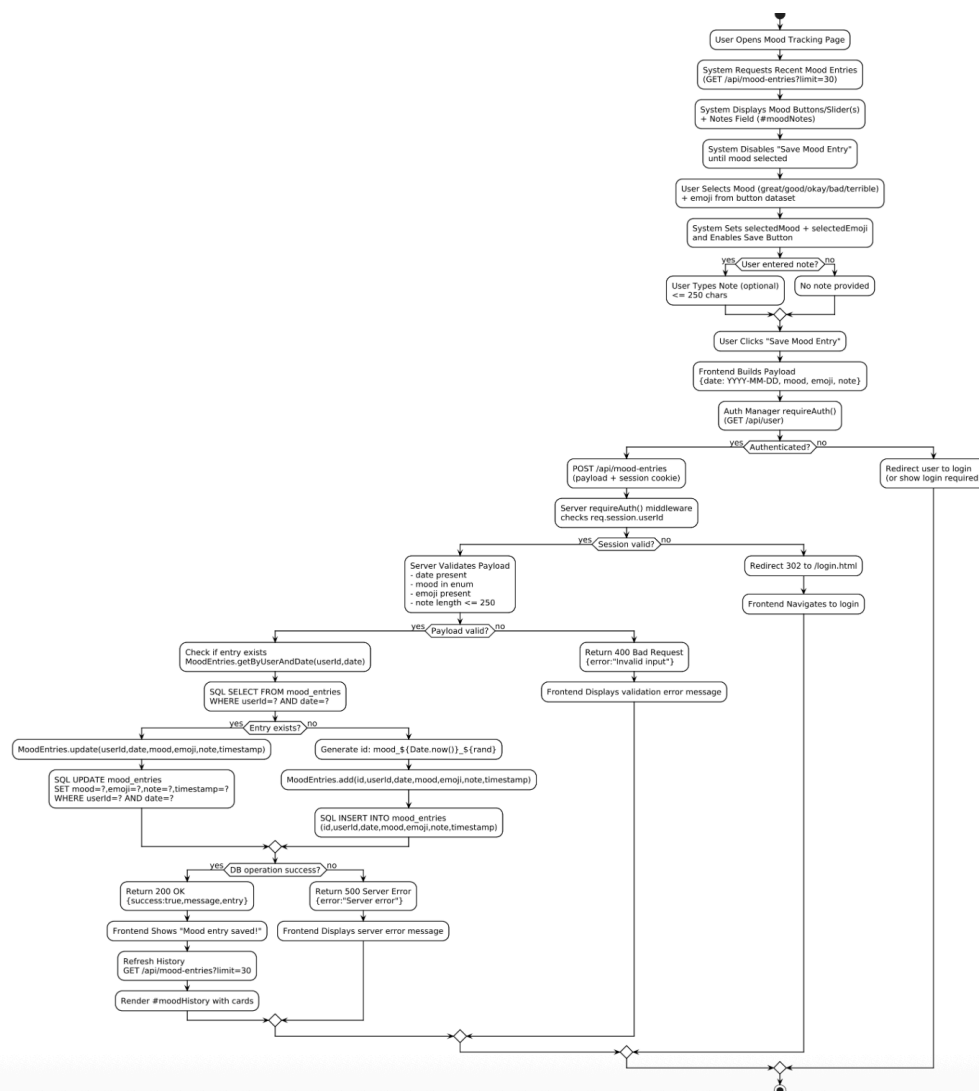


Wellness Tools Diagrams:

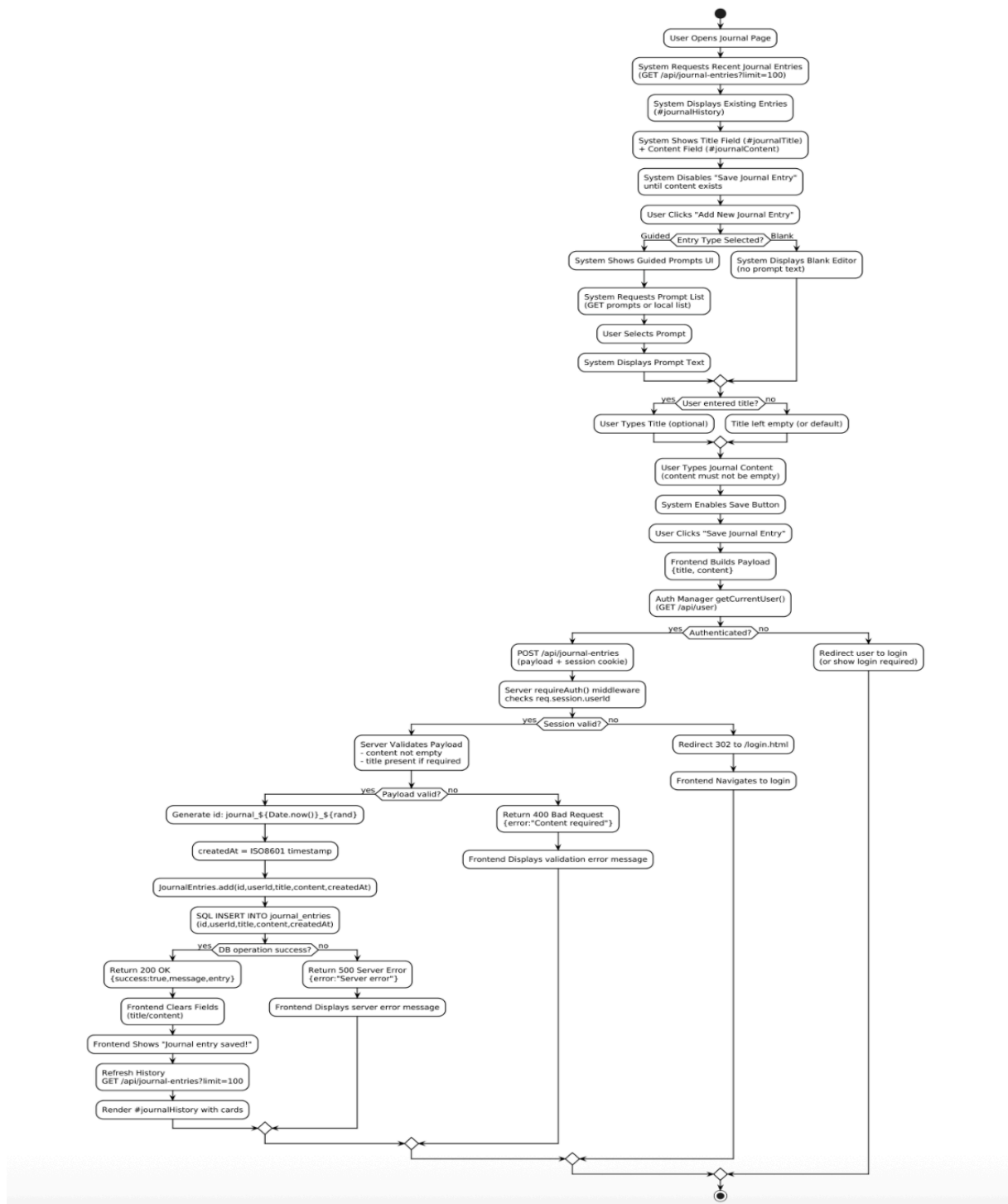


2.3 Sequence Diagram

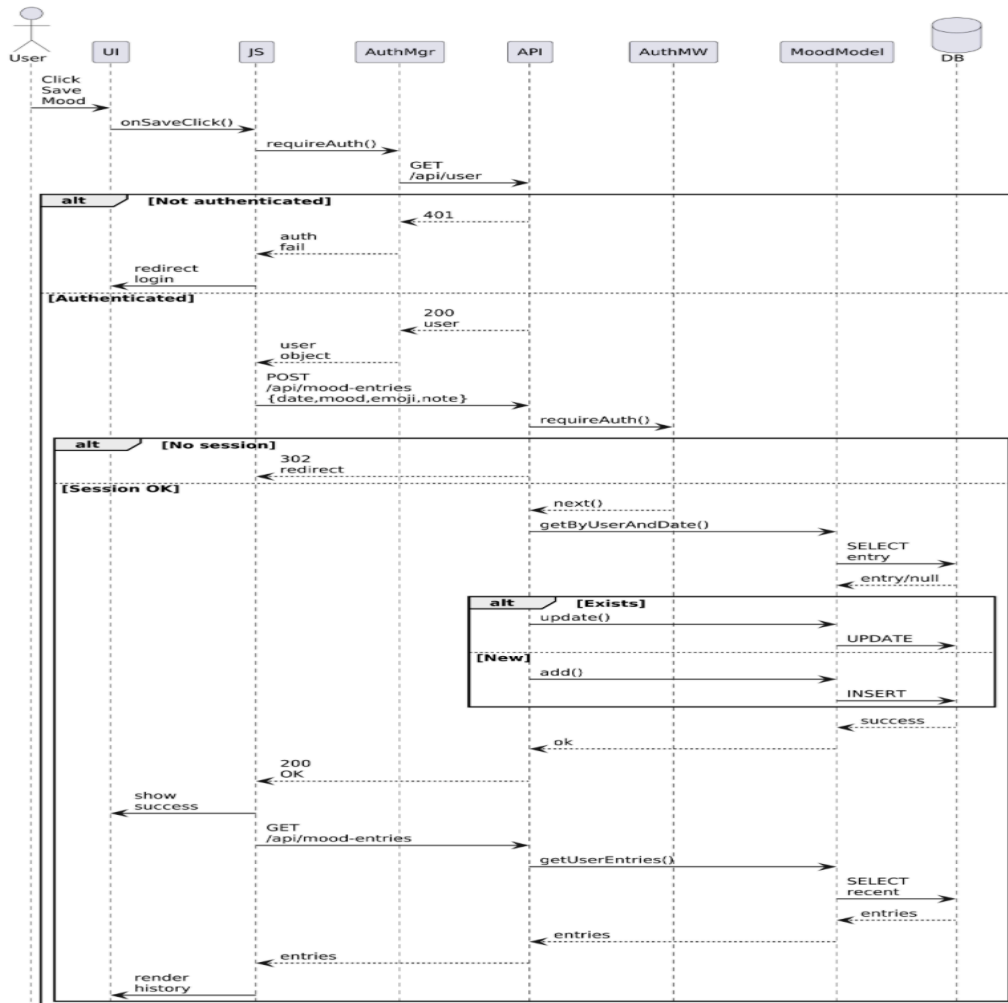
Mood Tracking:



Journal Page:



Select & Save Mood:



3. UPDATED BACKLOG WITH DESIGN STORIES

Week 4

DS-01: Define core domain model and class diagram

Design the core domain model and UML class diagram, including the following entities: User, MoodEntry, JournalEntry, Habit, HabitCompletion, GameScore, and ContentItem.

Dependencies: None

DS-02: Map database schema to domain model

Design the database schema and map it to the domain model, defining tables, primary keys, foreign keys,

constraints, and indexes.

Dependencies: DS-01

DS-03: Specify authentication flow and API contract

Define the authentication workflows and API contracts for user registration, login, logout, and retrieval of the current user.

Dependencies: DS-01

DS-04: Specify session management and route protection behavior

Design session handling and route protection logic, including redirects, guards, and access enforcement for protected routes.

Dependencies: DS-03

DS-05: Define role-based access control

Design role-based access control rules for administrative routes and corresponding UI gating behavior.

Dependencies: DS-03

DS-06: Design content management subsystem

Define the design of the content management subsystem, including the content.json schema, validation rules, and CRUD operations.

Dependencies: DS-01, DS-05

DS-07: Define mood tracking UX and data flow

Design the mood tracking user experience and data flow, including save/update logic, history retrieval, and input validation.

Dependencies: DS-01, DS-03

DS-08: Define habit lifecycle UX and data flow

Design the habit lifecycle workflow, including creation, completion, marking as done, and deletion.

Dependencies: DS-01, DS-03

DS-09: Define journal UX and data flow

Design the journaling workflows, including guided and free-entry modes, entry creation, and entry listing.

Dependencies: DS-01, DS-03

DS-10: Define game score workflow

Design the game score workflow, including saving scores, tracking personal bests, and updating leaderboards.

Dependencies: DS-01, DS-03

DS-11: Define admin user management UX and constraints

Design the administrative user management workflows, including constraints such as preventing self role-change or self-deletion.

Dependencies: DS-05

DS-12: Define client data isolation rules

Design client-side data isolation rules, including per-user storage keys and data cleanup on logout.

Dependencies: DS-03

Week 9**DS-13: Detailed UI flow map for mood tracker**

Create detailed UI flow diagrams for the mood tracker, including states, validation errors, and empty states.

Dependencies: DS-07

DS-14: Detailed UI flow map for habits and streaks

Design detailed UI flows for habit completion, streak calculation, reward logic, and failure cases.

Dependencies: DS-08
DS-15: Detailed UI flow map for journal Design detailed UI flows for journal prompt selection, save feedback, and entry history display. Dependencies: DS-09
DS-16: Define dashboard data aggregation design Design dashboard aggregation logic, including mood counts, habit counts, and “completed today” metrics. Dependencies: DS-07, DS-08
DS-17: Define habit reminder design Design the habit reminder system, including notification timing and user permission prompts. Dependencies: DS-08
DS-18: Define game hub UX and availability states Design the game hub user experience, including available and “coming soon” states. Dependencies: DS-10
DS-19: Define error handling and feedback patterns Design standardized error handling and user feedback patterns for all API interactions. Dependencies: DS-03
DS-20: Define audit logging requirements Design audit logging requirements for habit actions, including retention and storage policy. Dependencies: DS-08
DS-21: Define admin content editing UX and validation Design administrative content editing workflows, including validation rules for prompts, games,

and exercises.

Dependencies: DS-06

DS-22: Define journal edit and delete flows

Design journal entry edit and delete workflows, including confirmation dialogs and API usage.

Dependencies: DS-09

Week 12

DS-23: Verify design story alignment with implementation

Review all design stories against the current implementation to identify gaps and update acceptance criteria as needed.

Dependencies: DS-01 through DS-22

DS-24: Define test coverage targets for core flows

Define test coverage targets for authentication, mood tracking, habits, journaling, and games.

Dependencies: DS-03, DS-07, DS-08, DS-09, DS-10

DS-26: Finalize sequence and activity diagrams for critical use cases

Finalize sequence and activity diagrams for two critical use cases: authentication and habit completion.

Dependencies: DS-03, DS-08

DS-27: Finalize requirements-to-design traceability matrix

Create a traceability matrix mapping requirements to design stories and implementation tasks.

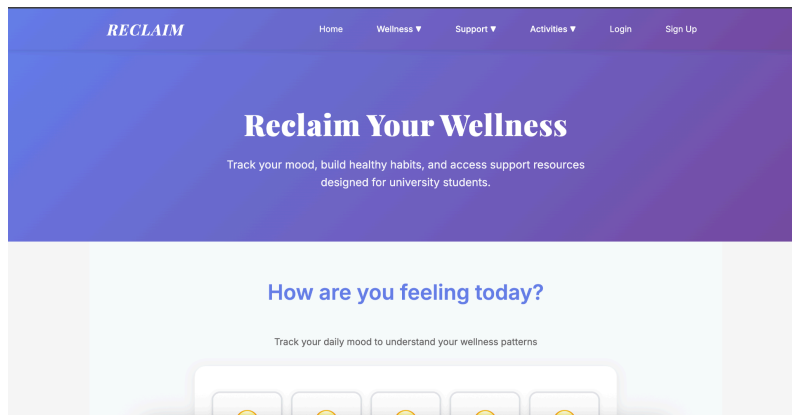
Dependencies: DS-01 through DS-26

4. GRAPHICAL USER INTERFACE DEVELOPMENT & TESTING

4.1 Screenshots of the initial GUI implementation

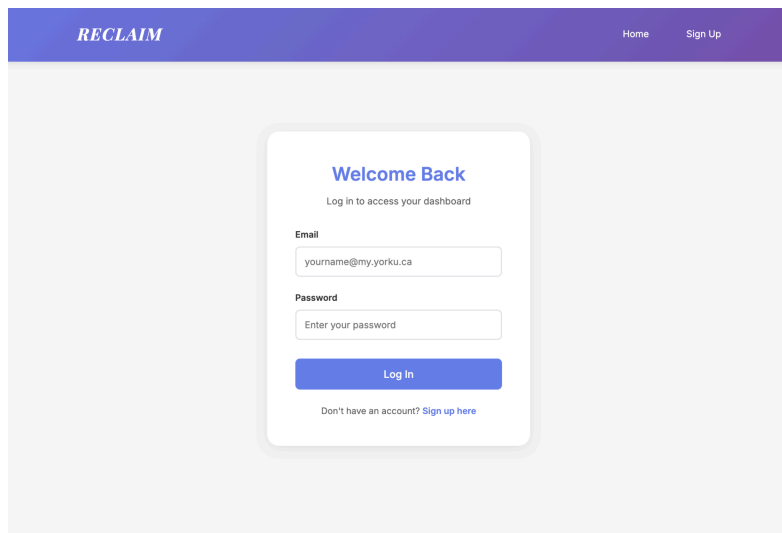
Home Page:

The home page highlights our mood tracker where they can track their mood of the day. All other features are locked until the user signs up or logs into their account.



Login:

The login card has a small shadow in order to focus the user's attention towards the card and logging into their account. Additionally, to further draw their focus, the navigation bar is shortened with "Home" and "Sign Up" being the only pages they can navigate to.



Sign Up:

The sign up card also uses similar design elements where there is a card with a shadow and a shortened navigation bar to draw the user's attention to the center of the page. In addition, to provide contrast of different visual elements, at the bottom of the sign up card, the "Note" is highlighted with a blue tab to match the colour scheme.

The screenshot shows a web page for 'RECLAIM' with a purple header containing 'Home' and 'Login' links. The main content is a white card titled 'Create Your Account' with the subtitle 'Join Reclaim to track your wellness journey'. The form includes fields for 'YorkU Email' (with placeholder 'yourname@my.yorku.ca' and a validation note), 'Password' (with a note about 8 characters and special characters), and 'Confirm Password'. A blue 'Create Account' button is at the bottom of the form, followed by a link 'Already have an account? Log in here'. A blue-bordered note box at the bottom states: 'Note: Admin accounts are created by developers. Only student accounts can be created through this form.'

Dashboard:

The dashboard is designed with cards for each feature and follows our design pattern. In addition, it includes a "Wellness Overview" that nicely displays the number of mood entries, active habits and completed today with a shaded rectangular background.

The screenshot shows a dashboard for 'RECLAIM' with a purple header containing 'Home', 'Wellness', 'Support', 'Activities', and 'Account' links. The main content is a white card titled 'Welcome to Your Dashboard' with the user email 'bnguyen@my.yorku.ca'. Below the title are three feature cards: 'Mood Tracker' (with a 'Track Mood' button), 'Habits' (with a 'View Habits' button), and 'Resources' (with a 'View Resources' button). Below these is a 'Habit Reminders' card with a 'Reminder time' input field and a 'Save Reminder' button. At the bottom is a 'Your Wellness Overview' card with three shaded rectangular boxes showing '0 Mood Entries', '0 Active Habits', and '0 Completed Today'.

Mood tracker:

The mood tracker contains a card to record your mood for the day and a card for each of your previous mood entries. When there are no previous entries, a card displays indicating that the user should add their first entry.

The screenshot shows the 'RECLAIM' website's mood tracker. The header is purple with the logo and navigation links: Home, Wellness, Support, Activities, and Account. Below the header is a purple banner with the text 'Reclaim Your Wellness' and a subtitle 'Track your mood, build healthy habits, and access support resources designed for university students.' The main content area is light blue and features a section titled 'How are you feeling today?' with the instruction 'Track your daily mood to understand your wellness patterns'. This section contains five mood selection buttons: 'Great' (happy face), 'Good' (neutral face), 'Okay' (neutral face), 'Not Good' (sad face), and 'Terrible' (very sad face). Below these buttons is a text input field labeled 'Add a note (optional)' with the placeholder 'What's on your mind today?'. A 'Save Mood Entry' button is at the bottom of this section. Below the mood entry section is a heading 'Recent Mood Entries' followed by a message: 'No mood entries yet. Start tracking your mood today!'.

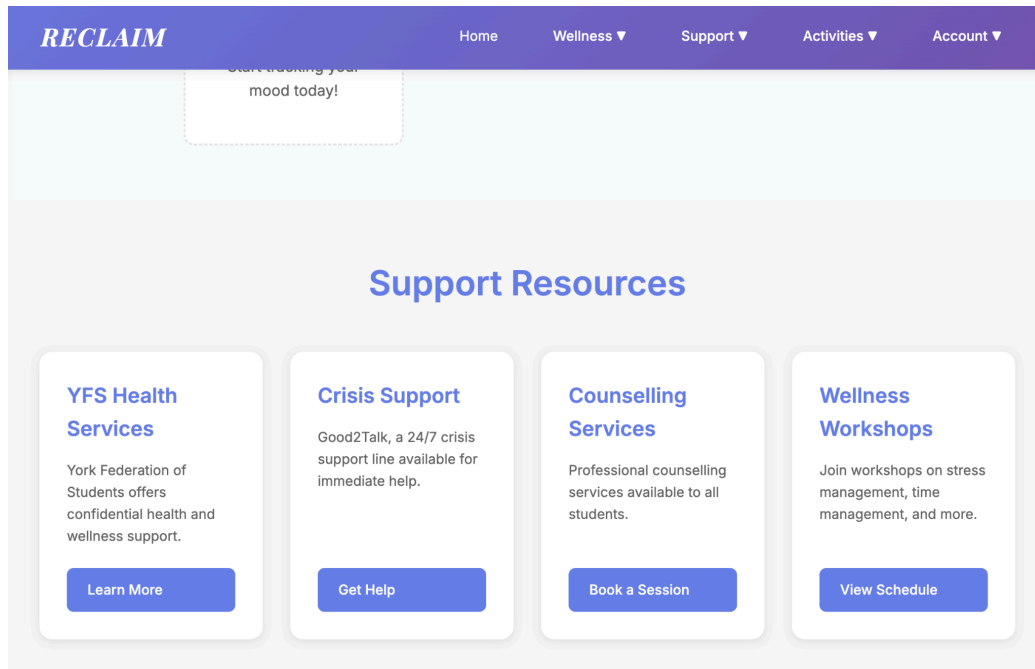
Habits:

The habits page contains a card to add a new habit and two sections beneath the current habits and the completed habits.

The screenshot shows the 'RECLAIM' website's 'Daily Habits' page. The header is purple with the logo and navigation links: Home, Wellness, Support, Activities, and Account. Below the header is a purple banner with the text 'Daily Habits' and a subtitle 'Build and track healthy habits to improve your wellness journey.' The main content area is light blue and features a section titled 'Create New Habit'. This section contains a form with three fields: 'Habit Name' (with a placeholder 'e.g., 30 minutes of exercise'), 'Description (Optional)' (with a placeholder 'Add details about this habit...'), and 'Reminder Frequency' (a dropdown menu set to 'Daily'). An 'Add Habit' button is at the bottom of this section. Below the 'Create New Habit' section is a heading 'Your Current Habits' followed by a message: 'No habits yet. Create your first habit above!'. Below this is another heading 'Completed Habits' followed by a message: 'No completed habits yet. Keep working on your goals!'.

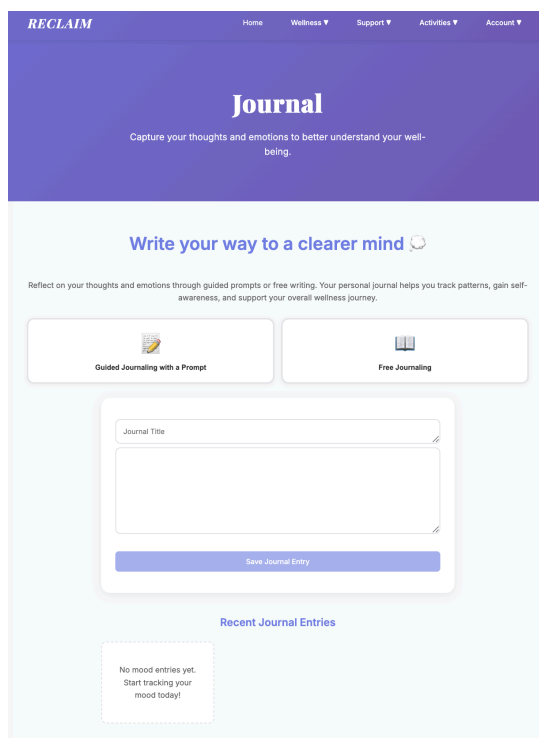
Resources:

The resources page contains four cards: YFS Health Services, Crisis Support, Counselling Services and Wellness Workshops. The button in each card corresponds to the call to action for that resource. For example, viewing the schedule of when the Wellness Workshops occur.



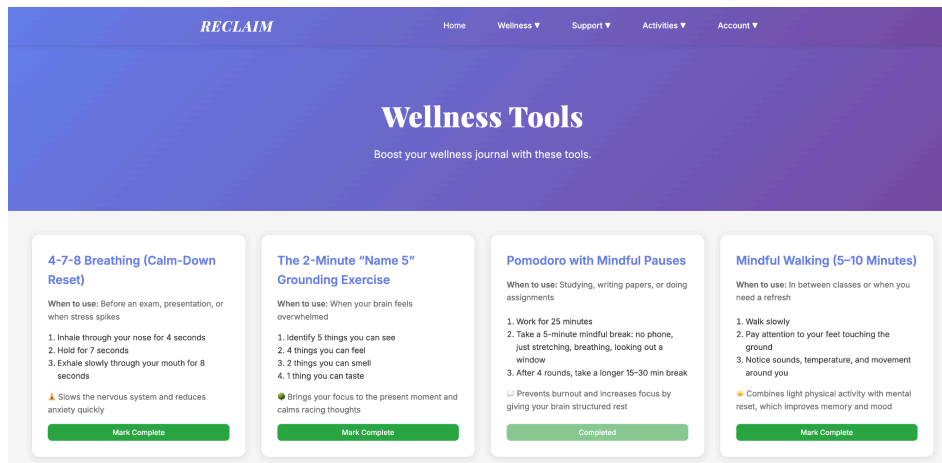
Journal:

The journal has two cards to start journaling: guided journaling with a prompt or free journaling. Underneath, there's a card for the user to input the title and body of their journal. At the end, they can view their recent journal entries.



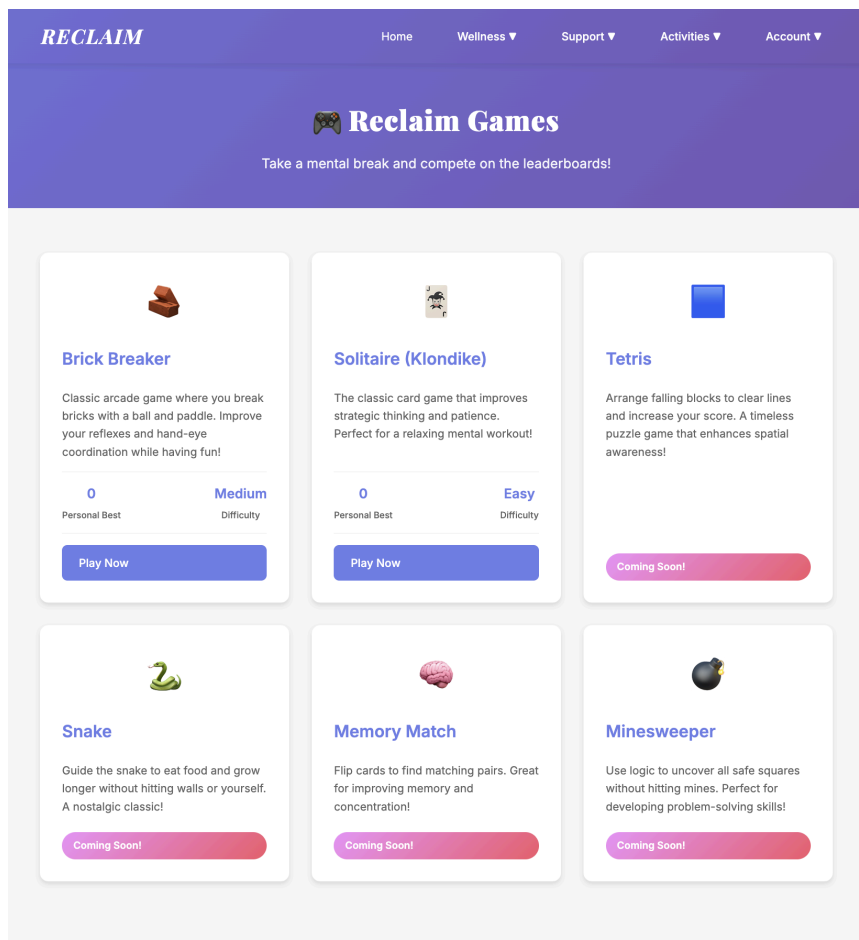
Wellness Tools:

The wellness tools contain four cards containing different strategies to mindfulness and better wellness. Once completed, the user can click the “Mark Complete” button for the associated card.



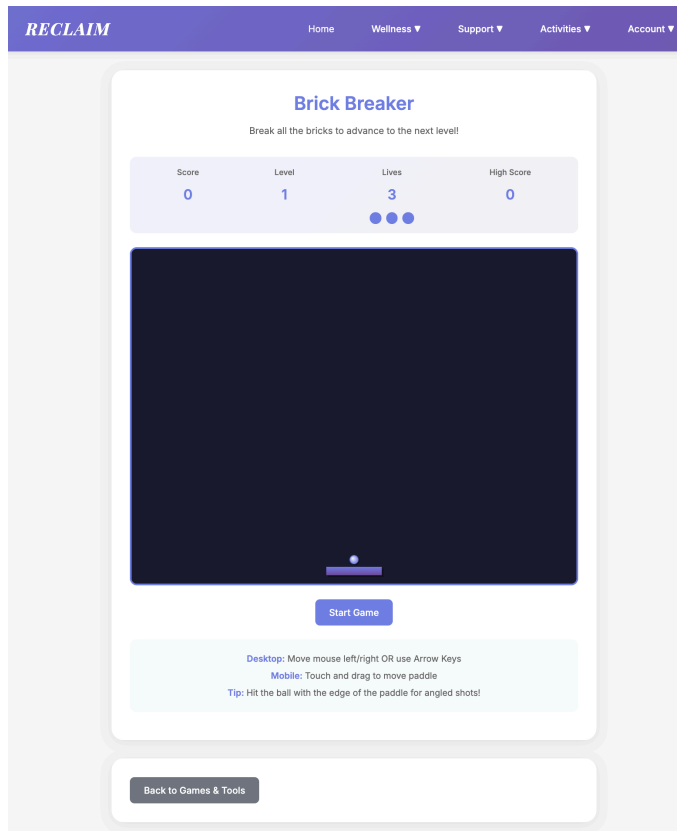
Games:

The games page contains six cards to display each game, two that are available and four that are coming soon. Each card contains a description of the game, and if it's available, a button to play the game, the user's personal best and difficulty level.



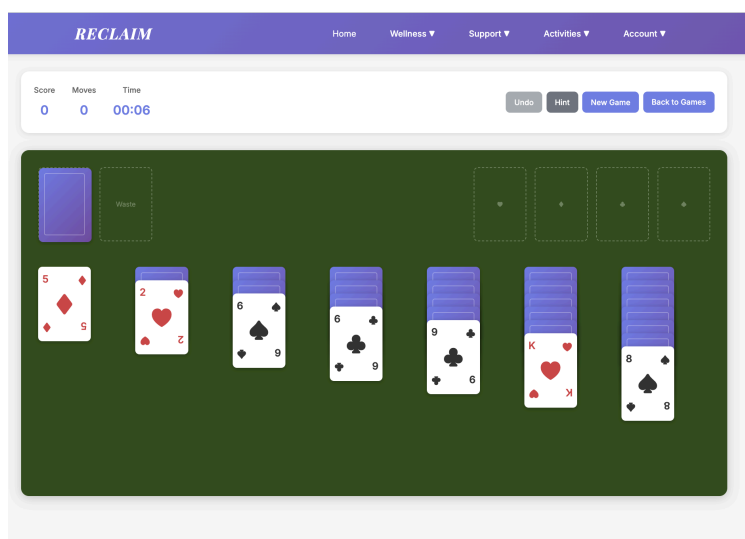
Brick Breaker (one of the games):

The brick breaker page contains one large card that contains the score, level, number of lives, high score, the game itself, the button to start the game and instructions to play the game on different devices. The bottom card contains the button to navigate back to the games page.



Solitaire (one of the games):

The solitaire page contains one card at the top with the score, moves and time. Additionally, the top card has various buttons such as undo, hint, new game and back to games. Beneath the top card, the user can play the solitaire game.



Manage content page (for admin):

The manage content page contains one large card to manage all content on the Reclaim application. Below the various buttons to manage the content, when a new user is added, a row is generated to input their role, email, etc.

The image displays two screenshots of the 'Manage Content' page in the RECLAIM application. Both screenshots feature a purple header with the 'RECLAIM' logo and navigation links: Home, Wellness, Support, Activities, and Account. The 'Account' dropdown menu is open in the top screenshot, showing options for Dashboard, Manage Content, and Logout.

The main heading is 'Manage Content', with the subtitle 'Add, edit, or delete prompts, games, and exercises'. Below this, there are four buttons: 'Journaling Prompts', 'Games', 'Exercises', and 'Manage Users'. A large blue button labeled 'Add Prompt' is visible in the top screenshot, while the bottom screenshot shows the 'Add User' button.

The bottom screenshot shows a table with two rows of user data. Each row includes an 'Email' field, a 'Role' dropdown menu (set to 'Admin'), a 'Created' date field, and 'Save Role' and 'Delete' buttons. The first row has the email 'r@my.yorku.ca' and the date '11/2/2025'. The second row has the email 'testadmin@my.yorku.ca' and the date '11/3/2025'. A note '(Cannot change your own role)' is present below the first row's role dropdown.

Email	Role	Created	Save Role	Delete
r@my.yorku.ca	Admin	11/2/2025	Save Role	Delete
testadmin@my.yorku.ca	Admin	11/3/2025	Save Role	Delete

4.2 Preliminary Testing and Test Coverage Report

Test Coverage:

File	% Stmts	% Branch	% Funcs	% Lines	Uncovered Line #s
All files	83.02	86.56	95.89	82.91	
Reclaim	82.51	84.58	92.5	82.33	
server.js	82.51	84.58	92.5	82.33	...-403,440,454-459,472-476,492
Reclaim/utils	100	100	100	100	
auth.js	100	100	100	100	
Reclaim/utils/db	82.09	96.96	100	82.09	
GameScores.js	76.92	100	100	76.92	27-28,44-45,63-64
Habits.js	80.32	100	100	80.32	34-35,51-52,80-81,109-110,126-1
JournalEntries.js	75.75	100	100	75.75	28-29,45-46,60-61,74-75
MoodEntries.js	75.75	100	100	75.75	28-29,43-44,60-61,74-75
Users.js	86.53	100	100	86.53	29,44,55,92-93,110-111
db.js	100	75	100	100	5
index.js	100	100	100	100	
Test Suites: 13 passed, 13 total					
Tests: 378 passed, 378 total					
Snapshots: 0 total					
Time: 30.458 s, estimated 31 s					
Ran all test suites.					
Force exiting Jest: Have you considered using `--detectOpenHandles` to detect async operations					

Overall, the test suite achieves 83% coverage. Several critical components are fully covered. The test suite achieves 83% overall coverage, with several critical components fully covered (100%), including the email validation logic in auth.js (notably YorkU domain verification) and database initialization and exports (db.js, index.js). Most of the remaining codebase maintains strong coverage (approximately 75–87%), encompassing all authentication, habits, mood tracking, journal, game score, and administrative API endpoints, along with database models, CRUD operations, queries, and user-isolation logic. Front-end coverage includes core GUI workflows such as login/logout, habit creation and display, and form interactions with validation feedback, giving confidence in key user flows.

The remaining ~17% gap stems primarily from code paths that are impractical to test under normal conditions. These include database failure catch blocks that only trigger during unexpected SQLite errors, requiring heavy mocking or forced corruption, and server startup logic (app.listen), which runs only in production and is intentionally excluded from test execution.

Tools used

New:

- **Jest:** A JavaScript testing framework for unit and integration tests. Provides test runner, assertions, mocking, and code coverage via Istanbul
- **@playwright/test:** End-to-end browser testing framework. Automates Chromium and Firefox to test real user interactions with the GUI.
- **Supertest:** HTTP assertion library for testing Express endpoints. Allows making requests to the app without starting a server.
- **Cross-env:** Cross-platform environment variable setting. Ensures NODE_ENV=test works on Windows, Mac, and Linux.

Already existing:

- **Better-sqlite3:** Database driver - tests use same DB module as application

- **Bcrypt:** Password hashing - tests verify hashed passwords work
- **Express-session:** Session management - tests verify authentication state